

NeuroX Adaptive

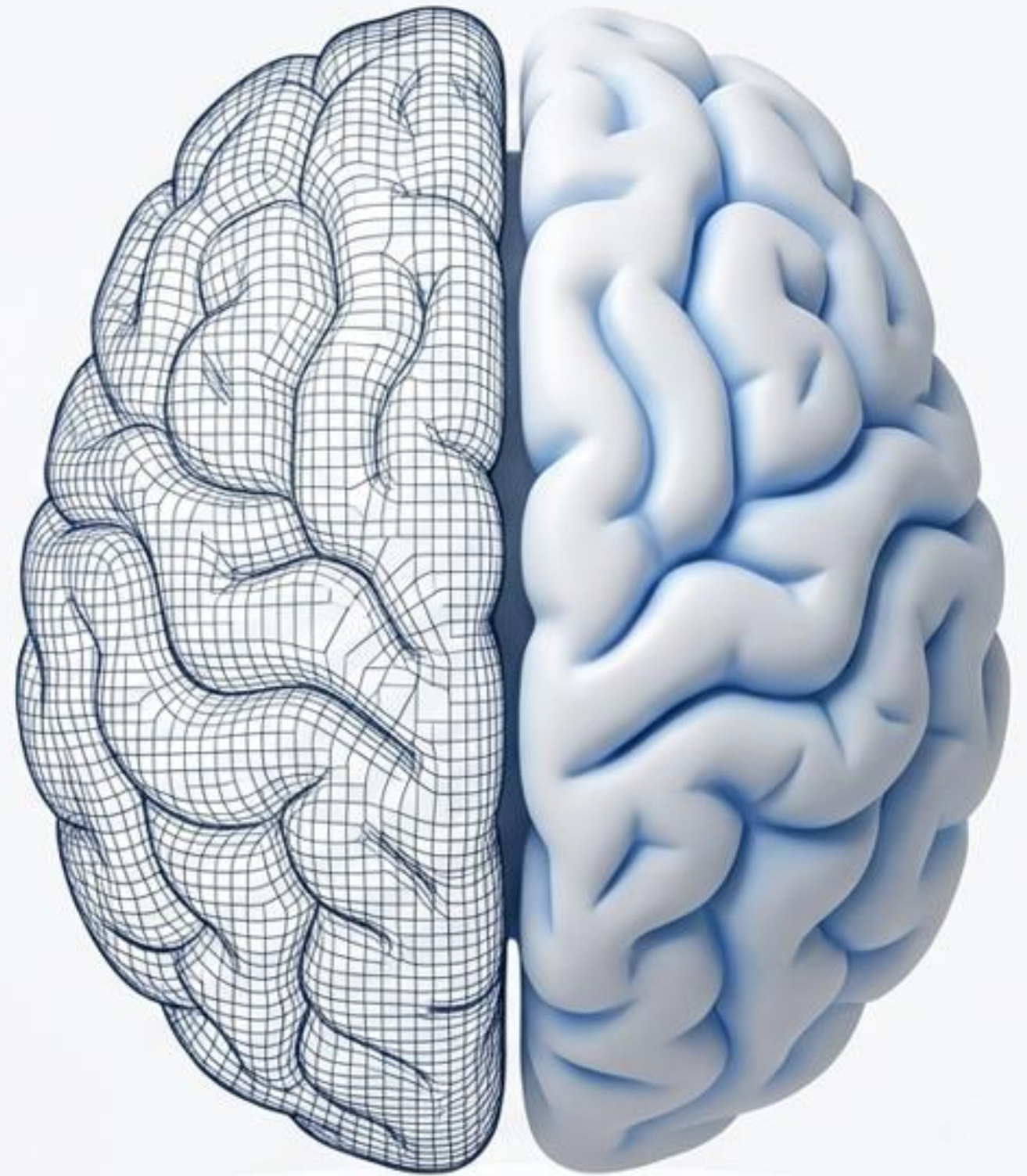
A Hybrid 3D CNN-Transformer
Framework for Holistic
Neuroimaging

Team Members:

1. Sandireddy Manideep (22BCE1434)
2. Munta Sai Karthik (22BCE5060)
3. Gotam Sai Varshith (22BCE1605)

Guide Name:

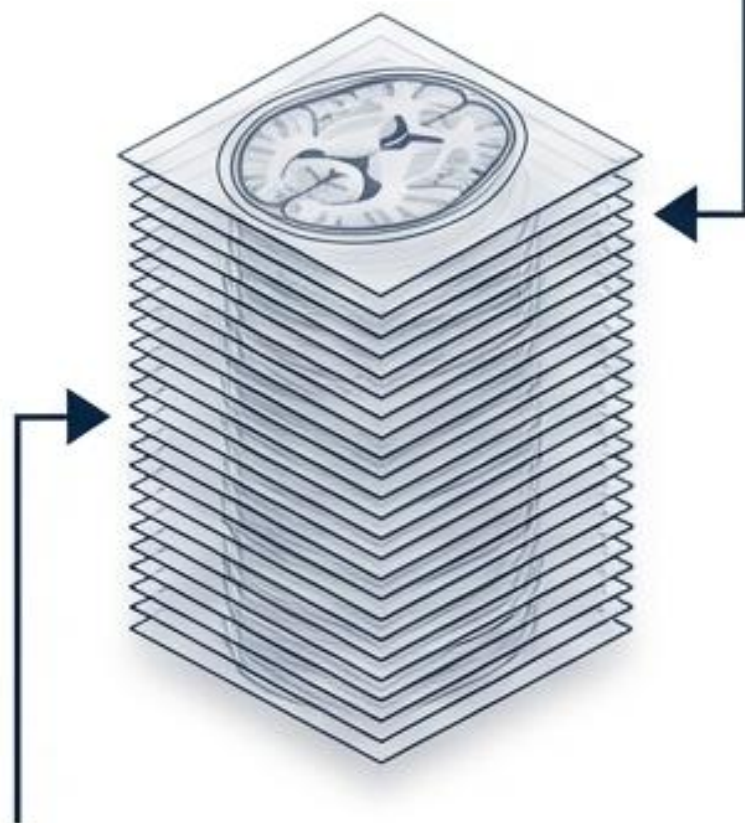
1. Dr. Jayaram B



Introduction

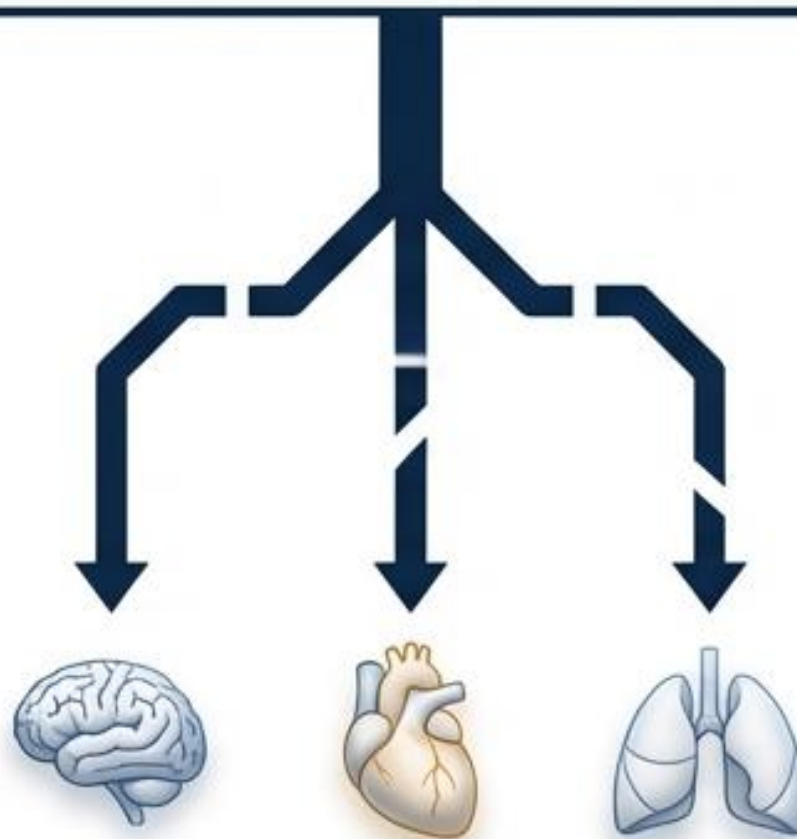
- **Current Diagnostic Bottleneck:** Modern medical imaging, such as MRI and CT, produces massive 3D volumetric datasets (NIfTI) that require exhaustive manual review by radiologists, leading to potential diagnostic fatigue and delays.
- **The Problem of "Siloed" AI:** Existing Computer-Aided Diagnosis (CAD) tools are typically task-specific, meaning a clinician must run separate models for tumor detection, stroke analysis, and neurodegeneration (Alzheimer's).
- **The Need for Volumetric Context:** Standard 2D slice-by-slice analysis often fails to capture the global structural symmetry and long-range anatomical relationships necessary for early-stage pathology detection.

The Clinical Paradigm & Problem Statement



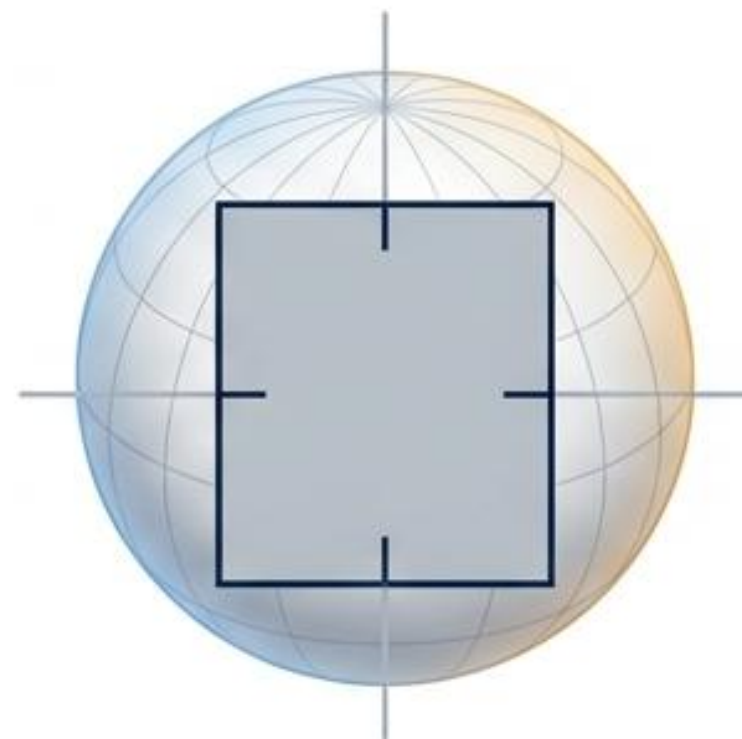
The Overload

Modern scans produce **hundreds** of slices per patient, directly causing severe radiologist fatigue.



The Fragmentation

Clinicians are forced into task-**specific** silos, running separate software for tumors, strokes, and Alzheimer's.



The Blindspot

Single-slice 2D analysis fundamentally misses volumetric depth and systemic structural symmetry.

Project Vision & Core Objectives

NeuroX Adaptive Core System

```
graph TD; A[NeuroX Adaptive Core System] --> B[Unification]; A --> C[Accuracy]; A --> D[Explainability]; A --> E[Automation];
```

Unification

1 Model, 1 NIfTI Volume. Simultaneous diagnosis of Brain Tumors, Stroke, and Alzheimer's.

Accuracy

Targeting >85% baseline precision leveraging a unique Hybrid 3D CNN-Transformer architecture.

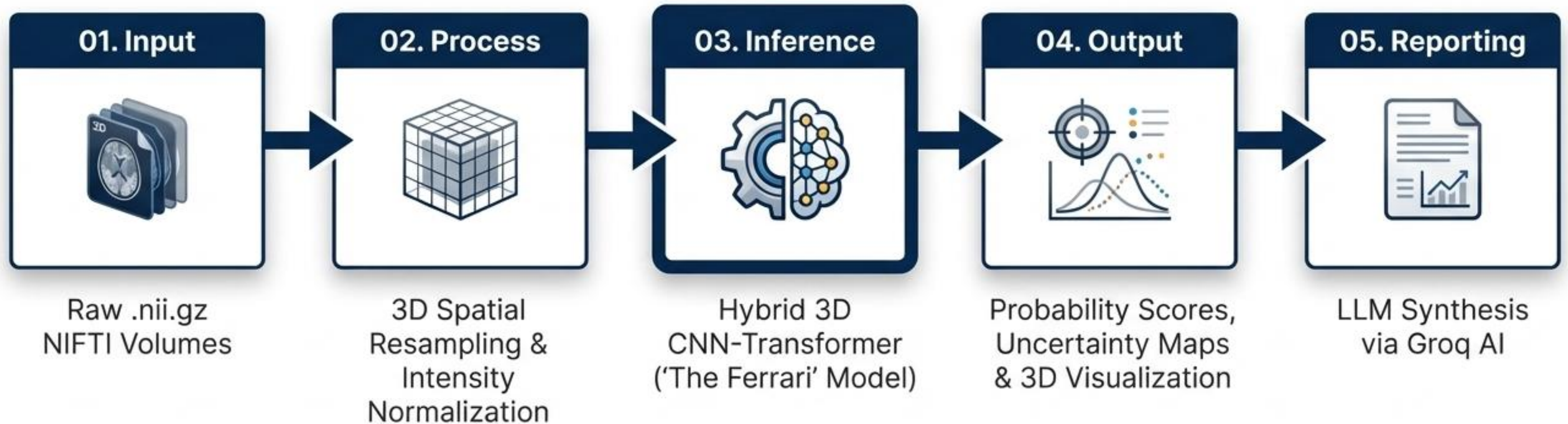
Explainability

Bridging the physician trust gap with 3D Grad-CAM heatmaps for instant visual verification.

Automation

Instant generation of structured radiology reports via Groq AI LLM synthesis.

High-Level System Workflow

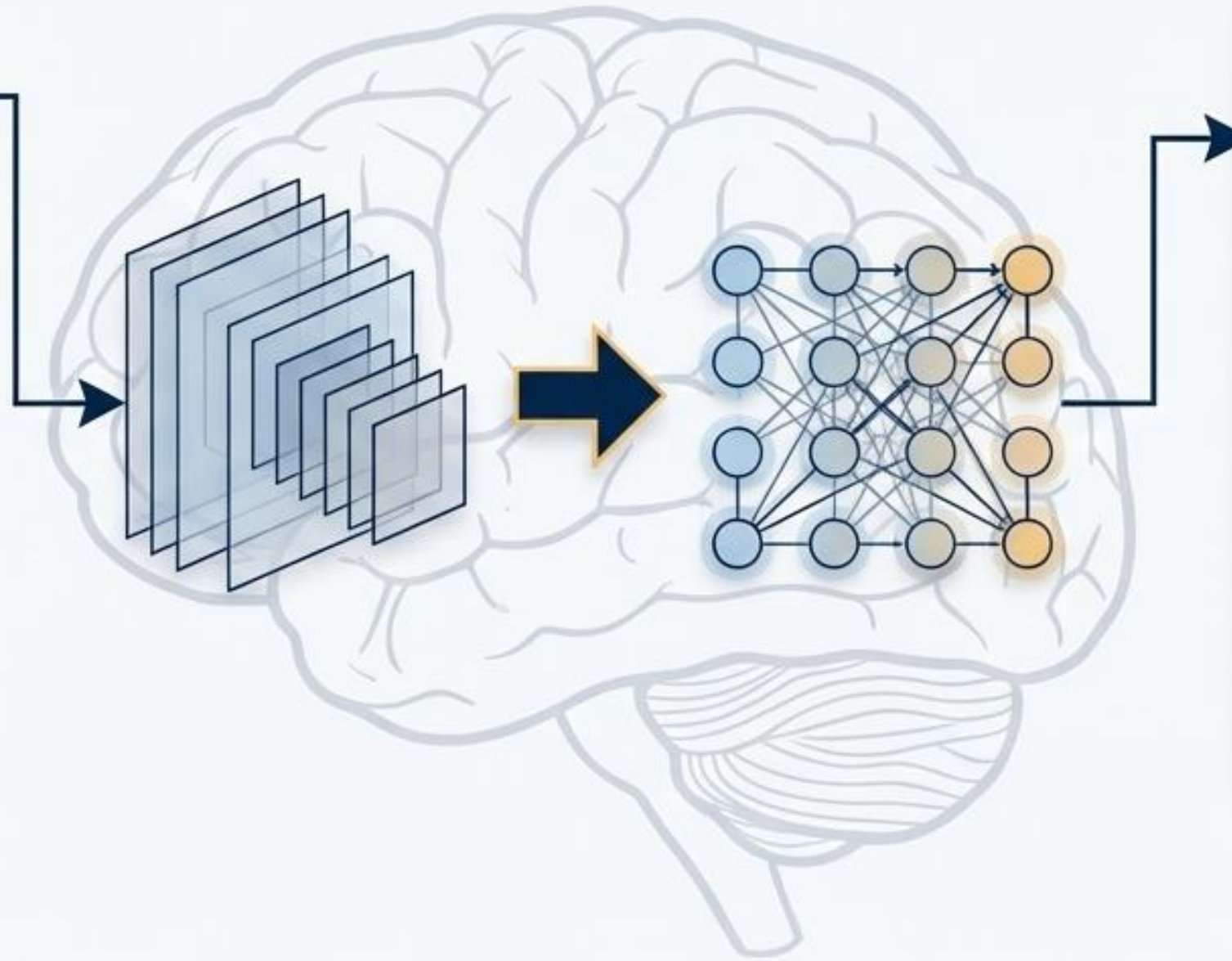


Core Architectural Innovation: The Hybrid Model

The Eyes: 3D CNN Encoder

Extracts local textures (e.g., connucter)

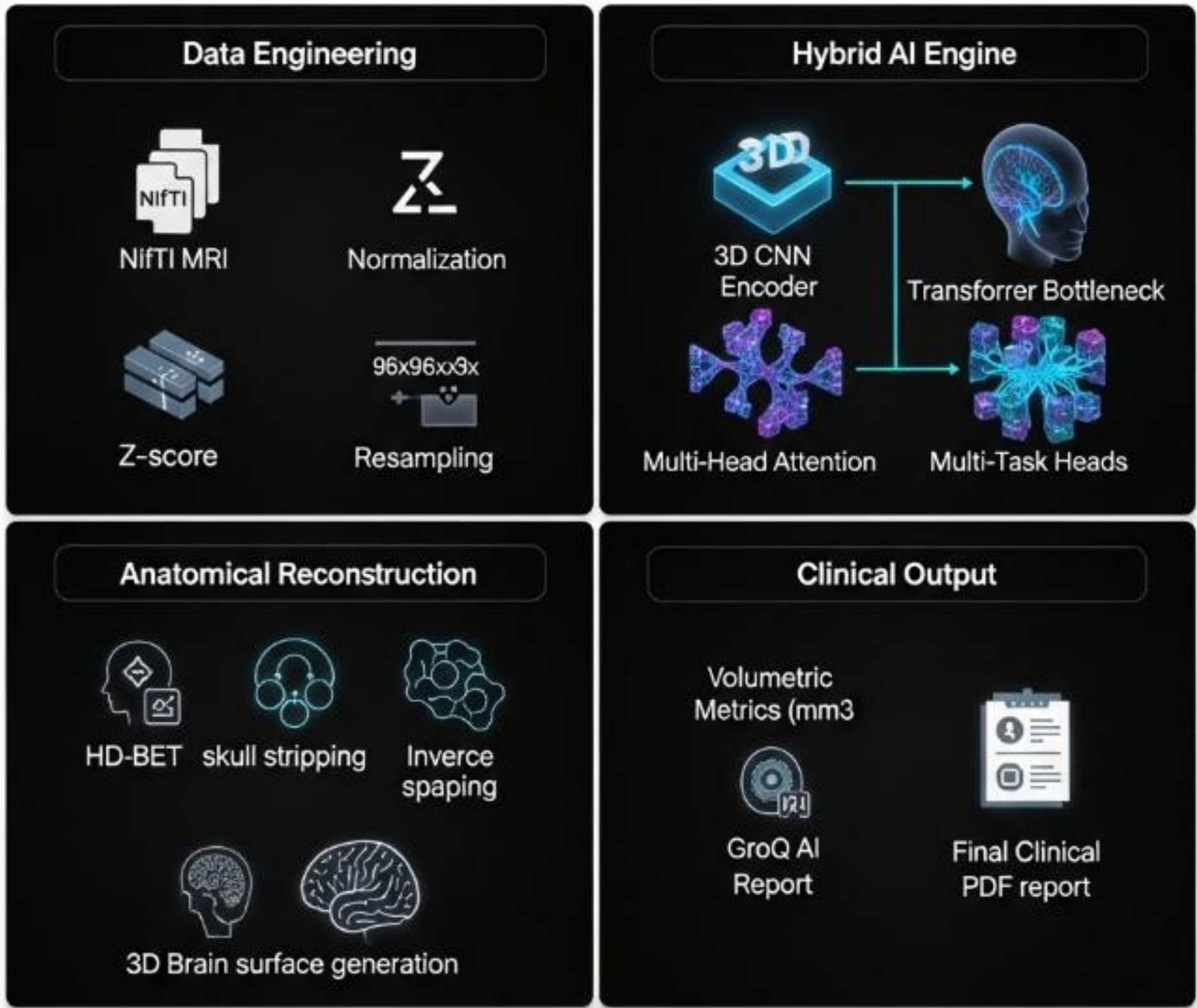
Extracts local textures (e.g., ragged tumor borders) using 3D convolutional kernels (3x3x3) and ResNet-style skip-connections.



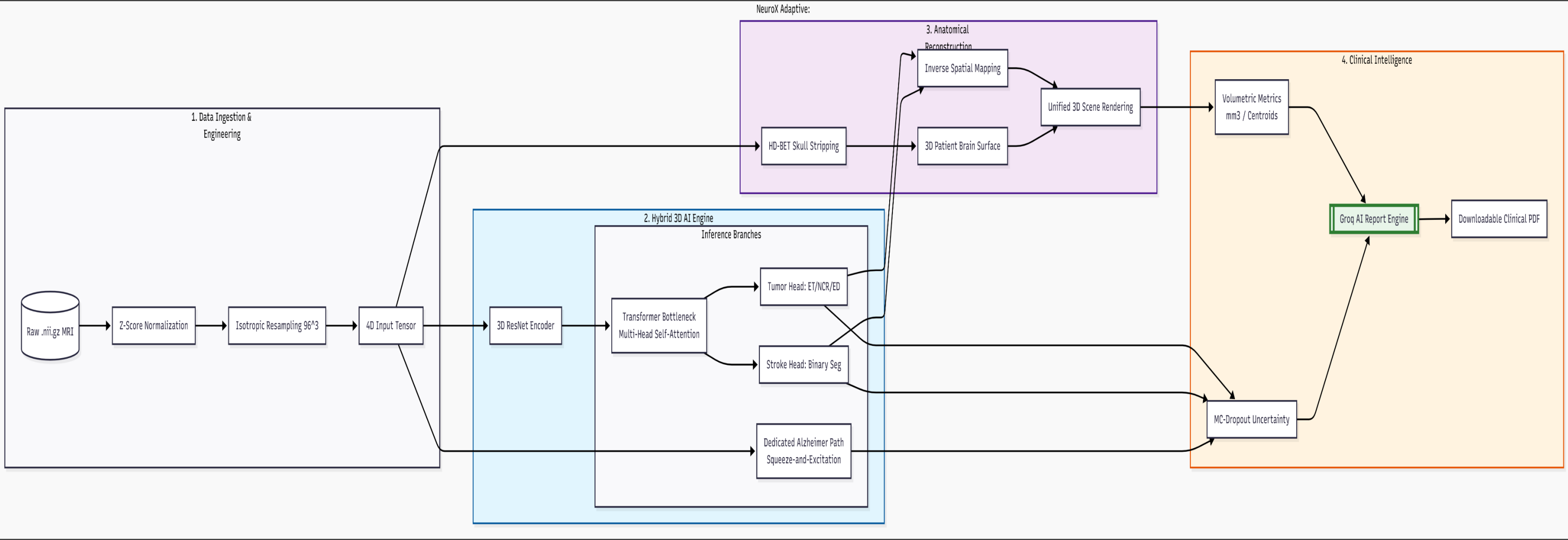
The Brain: Transformer Bottleneck

Maps global context. Employs Multi-Head Self-Attention and precise spatial encoding to understand long-range anatomical relationships across hemispheres.

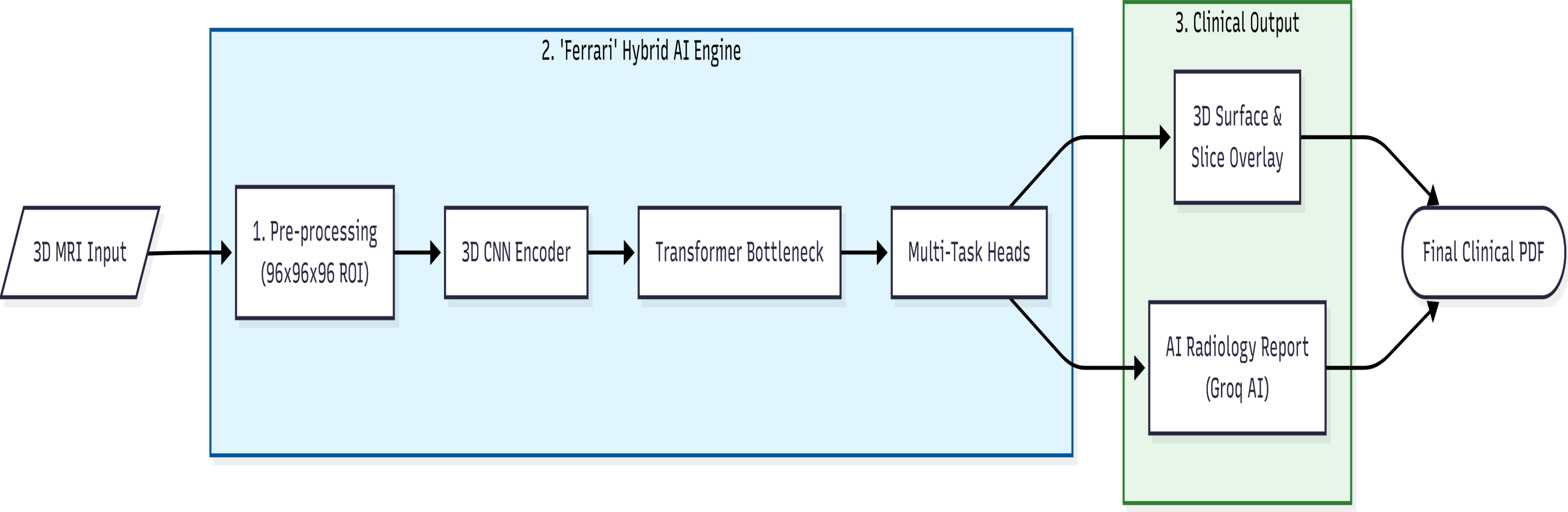
NeuroX Adaptive: Global Functional Architecture



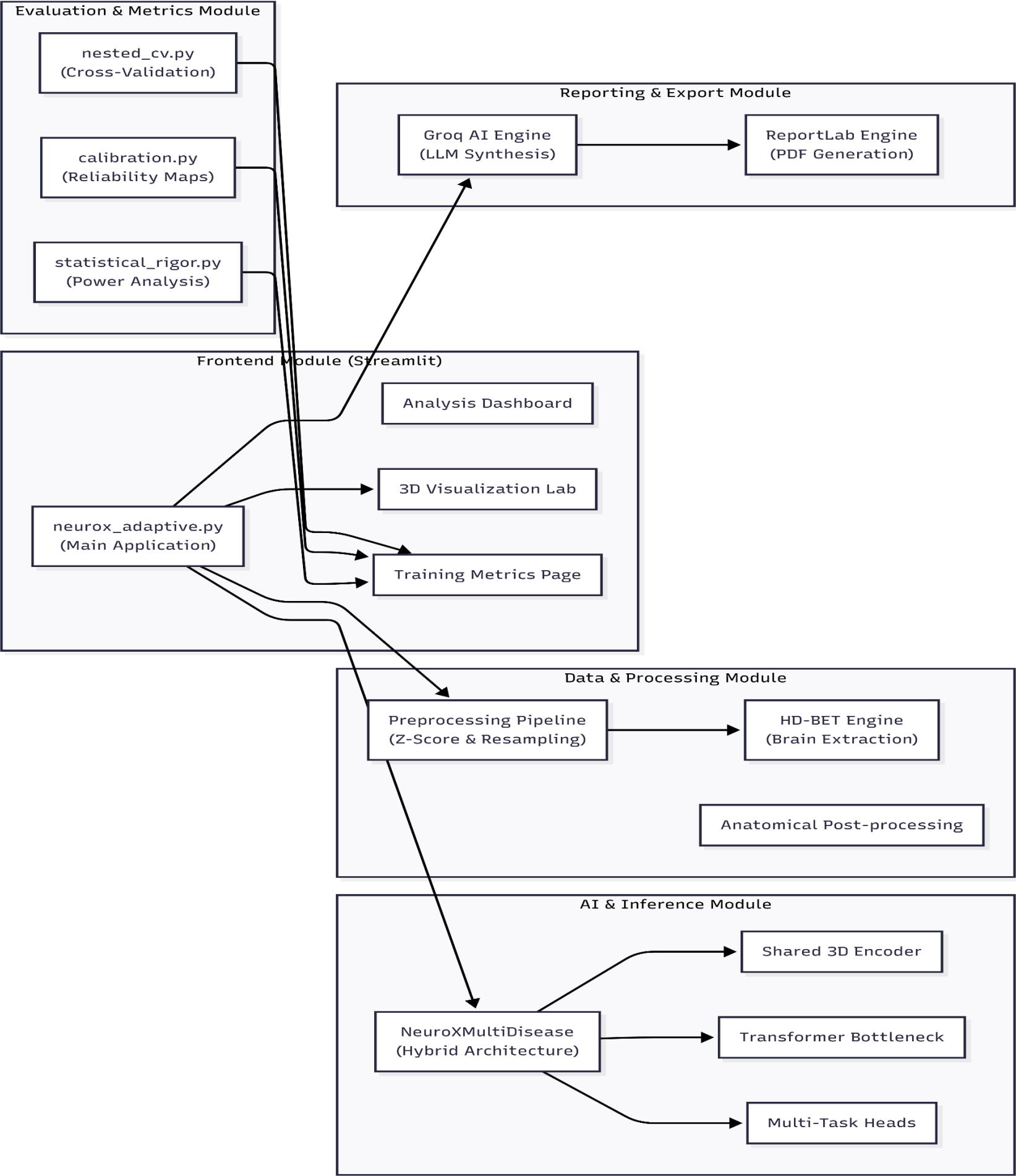
NeuroX Adaptive: Architecture Diagram



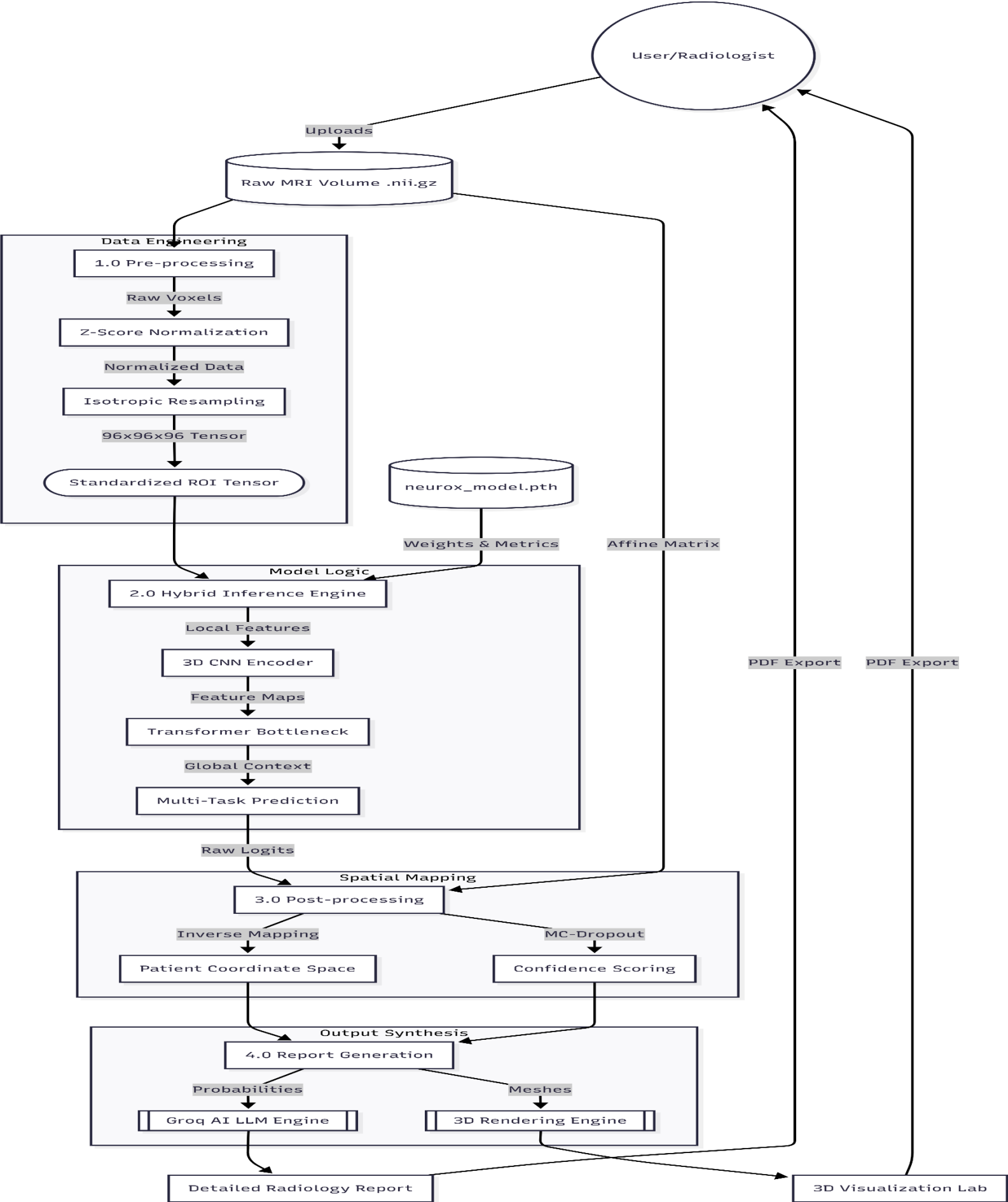
NeuroX Adaptive: Flow Chat



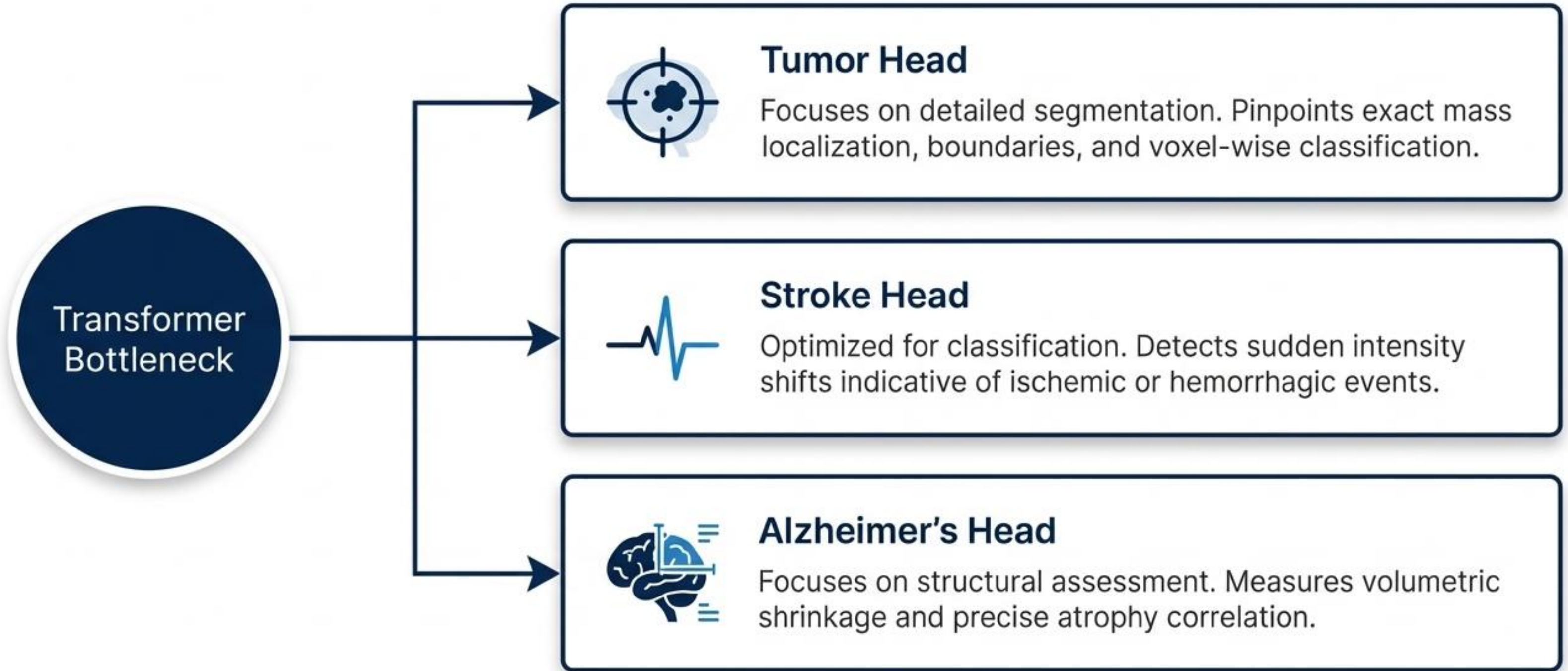
NeuroX Adaptive: Modal Diagram



NeuroX Adaptive: Data Flow Diagram

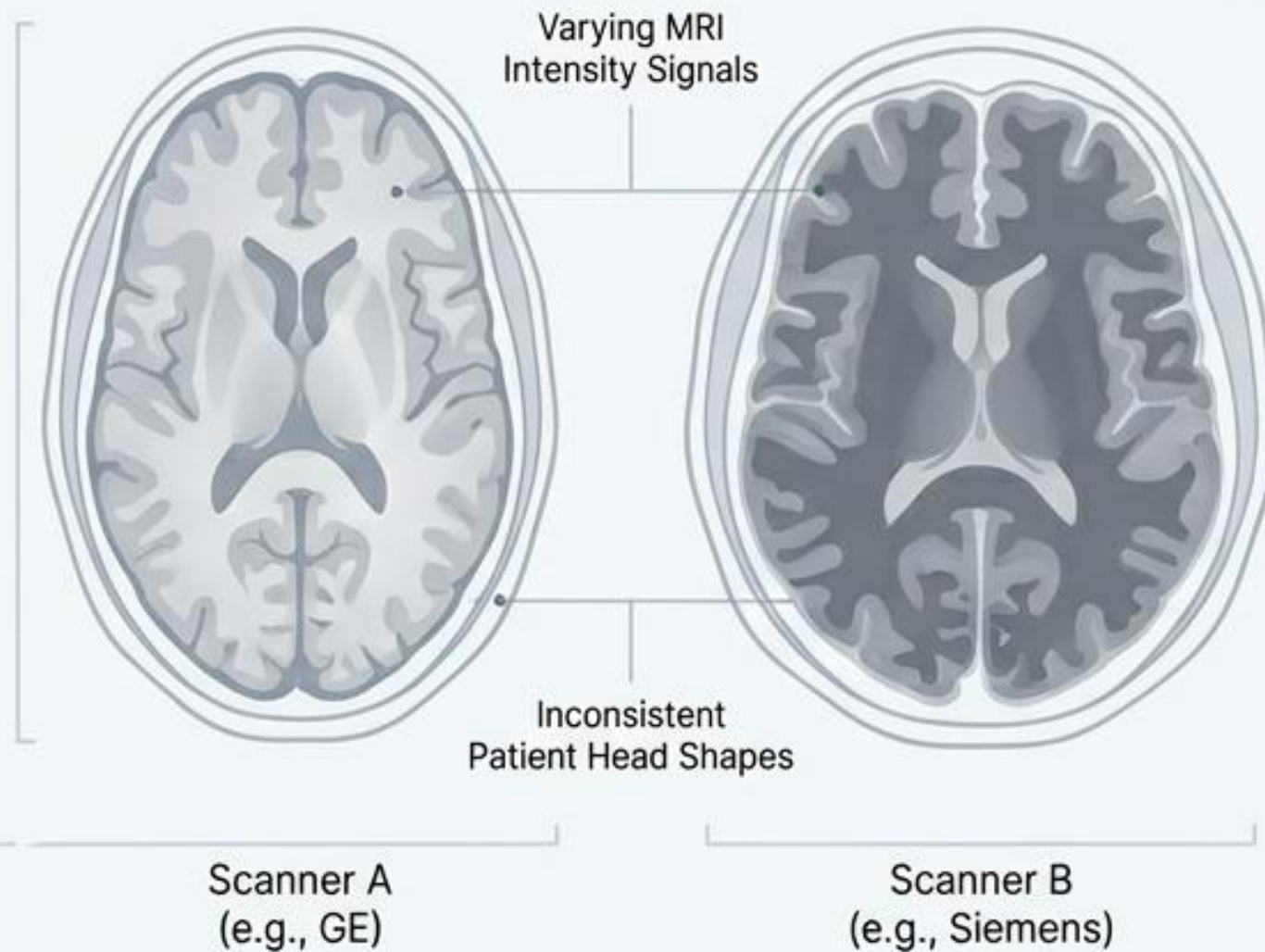


Multi-Task Decoupled Heads

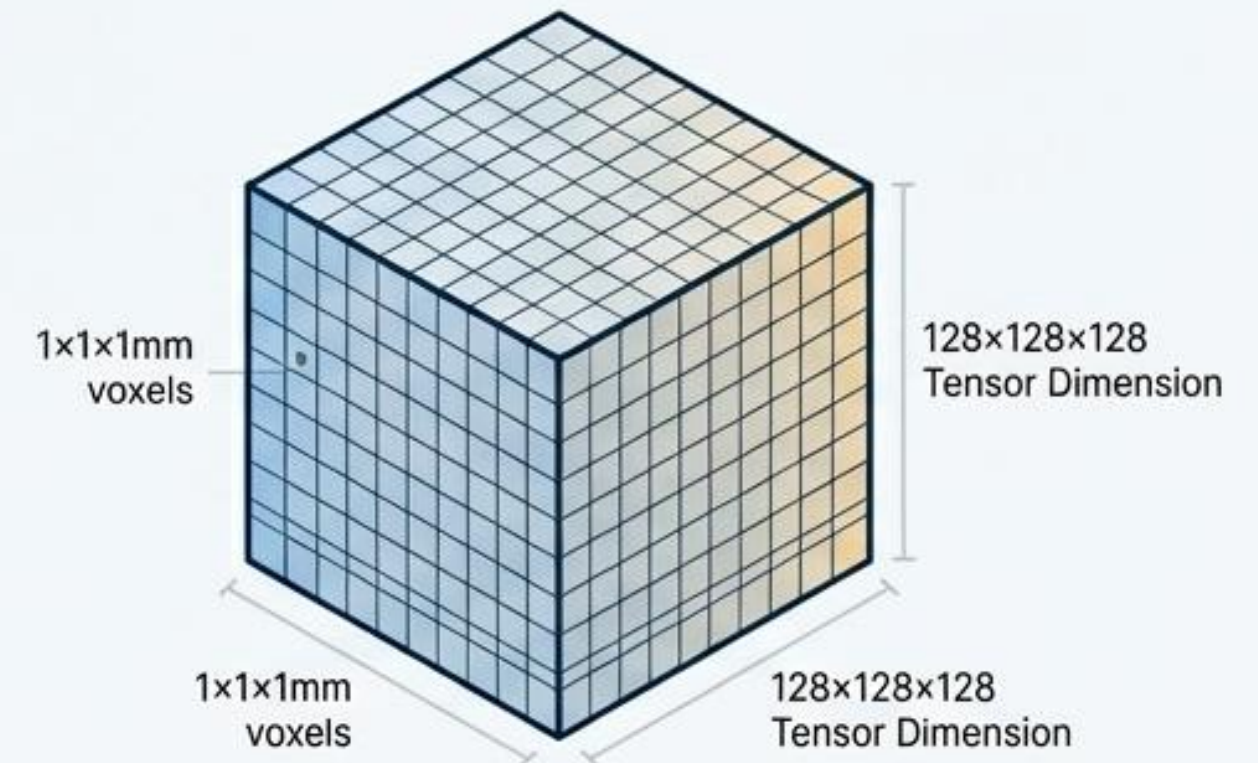


Data Engineering: The NIfTI Processing Pipeline

Raw Hardware Variance



Standardized Isotropic Grid



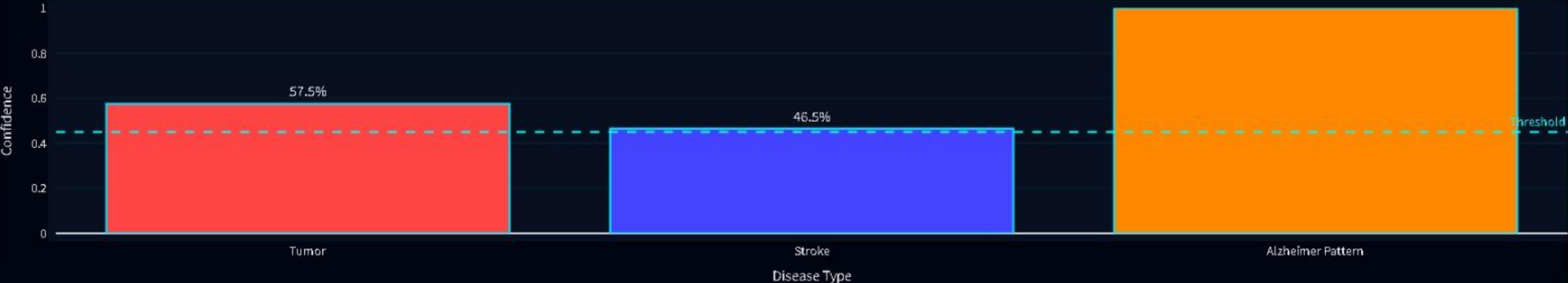
1. **Intensity Normalization:** Z-score & Min-Max Scaling standardizes white matter values across all hardware variations.
2. **Spatial Isotropic Resampling:** Forces a universal 1x1x1mm voxel grid, strictly cropped/padded to exactly 128x128x128 tensors.

Result:

Clinical Interpretation

- Confidence: 100.0% - Model's belief that alzheimer pattern is present
- Uncertainty: 0.000 - Model's uncertainty about this prediction
- Status: DETECTED - Based on 45% threshold
- Recommendation: Expert review recommended for confirmation

Disease Presence Confidence



Result:

🚨 Detected Pathologies

Tumor

Confidence: 57.5%

- 🔪 Volume: 12,912.6 mm³ (12.91 mL)
- 📍 Centroid: (56.6, -60.2, -50.6) mm
- 📏 Bounding Box (Min): (31.1, -91.7, -65.2) mm
- 📏 Bounding Box (Max): (103.0, -38.1, 6.9) mm

Stroke

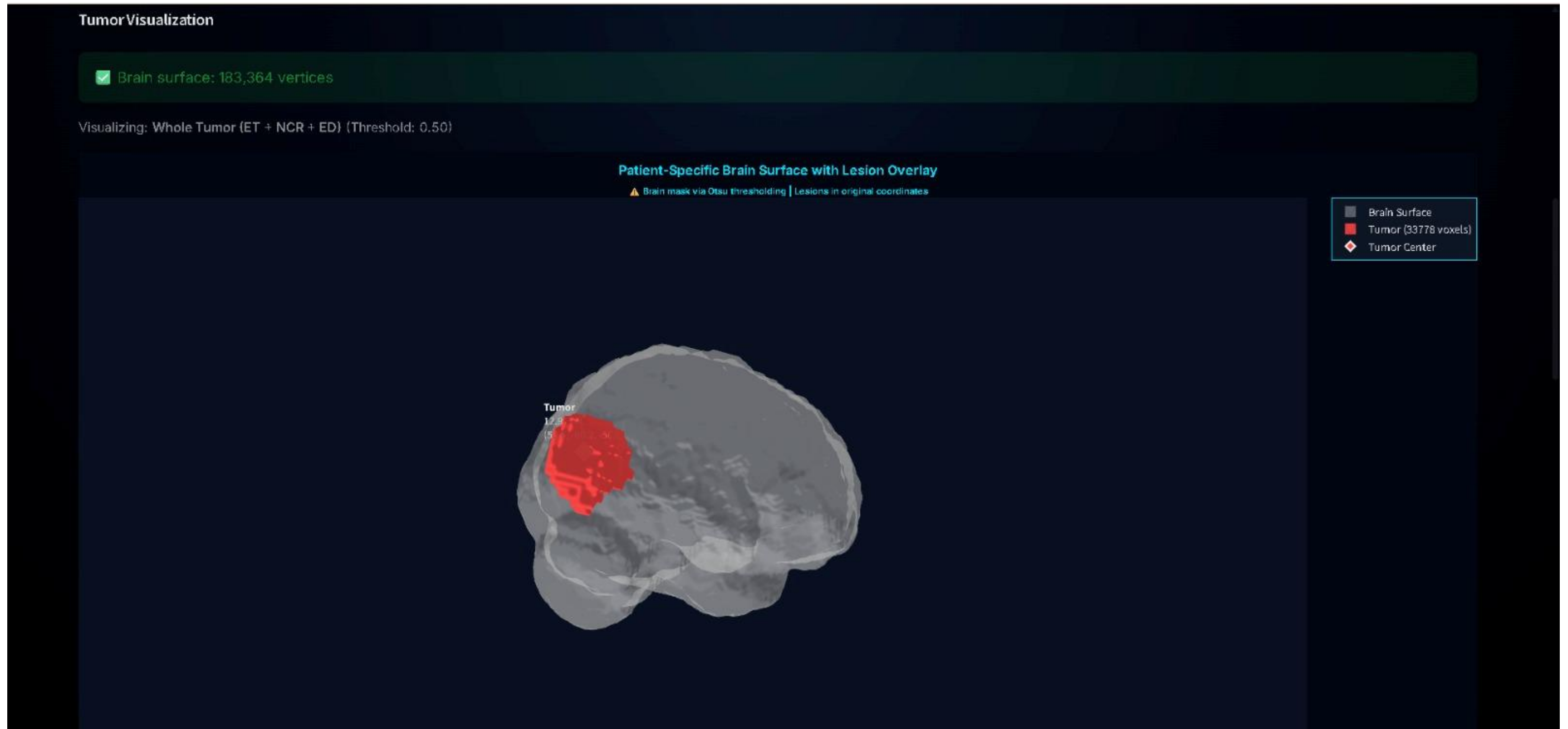
Confidence: 46.5%

- 🔪 Volume: 22,116.2 mm³ (22.12 mL)
- 📍 Centroid: (55.5, -20.2, -37.4) mm
- 📏 Bounding Box (Min): (32.4, -55.6, -59.8) mm
- 📏 Bounding Box (Max): (100.9, 34.3, 7.3) mm

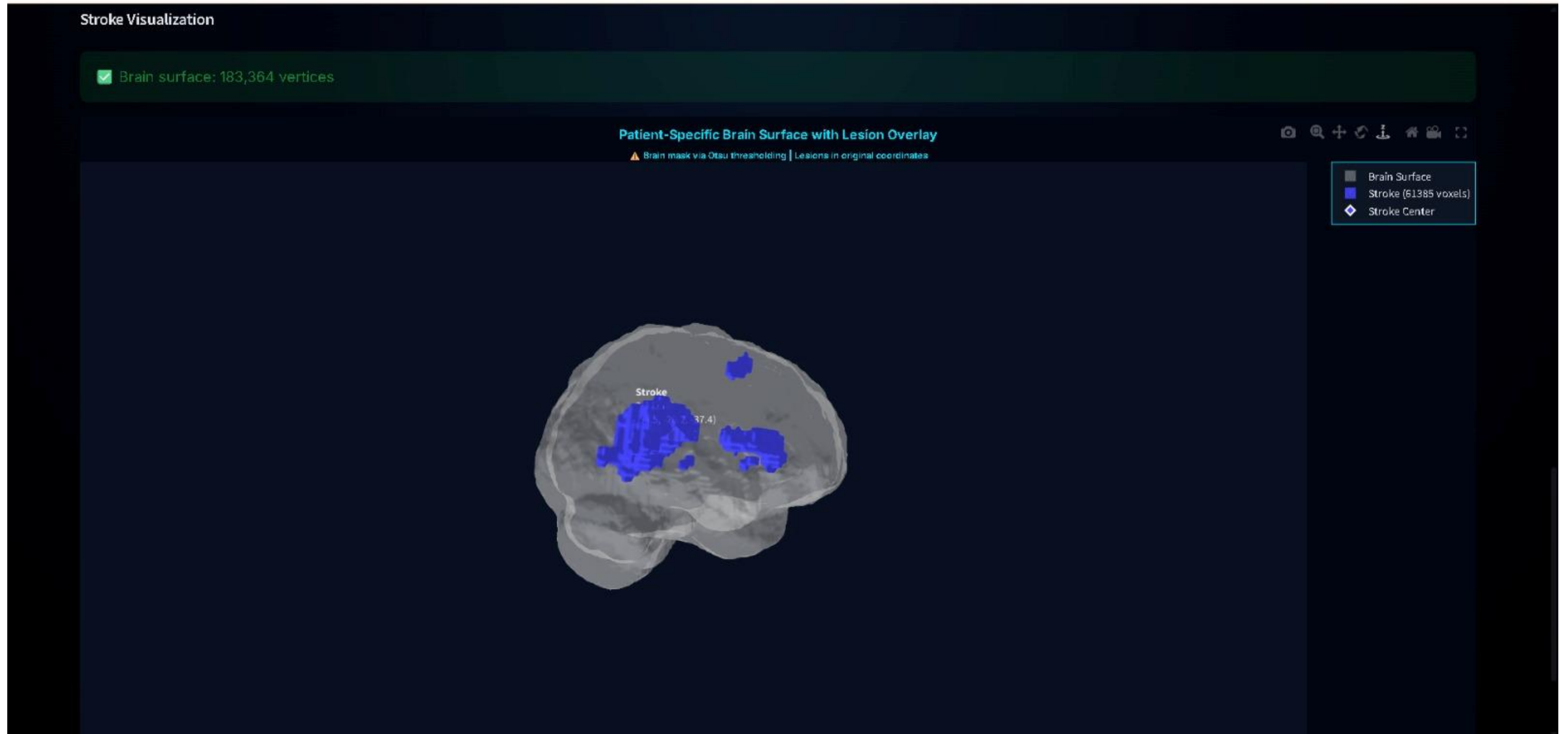
Alzheimer Pattern

Confidence: 100.0%

Result:



Result:



Result:



Result:



Result:

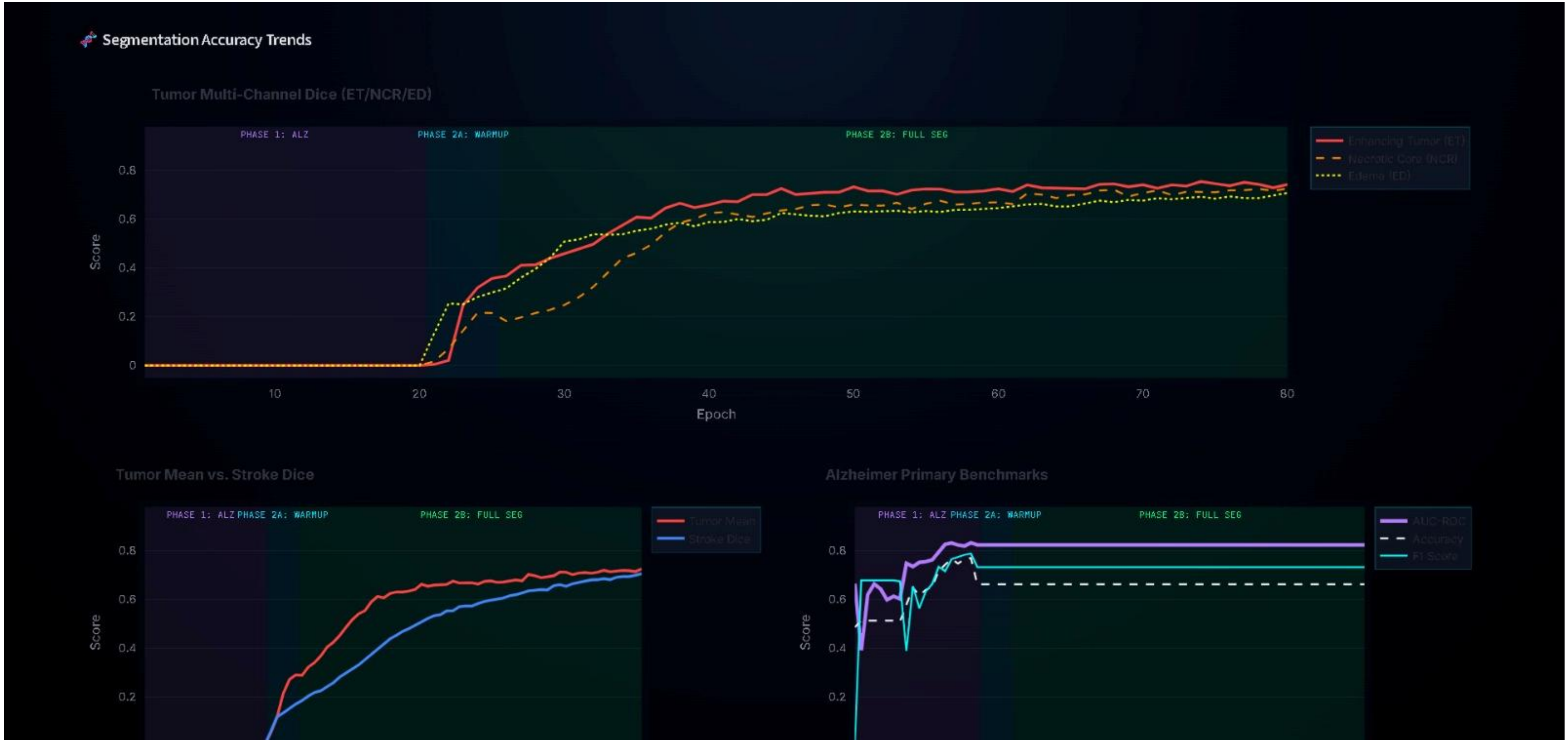
🏆 Detection Rigor (Alzheimer)



💡 Advanced Curriculum Analysis

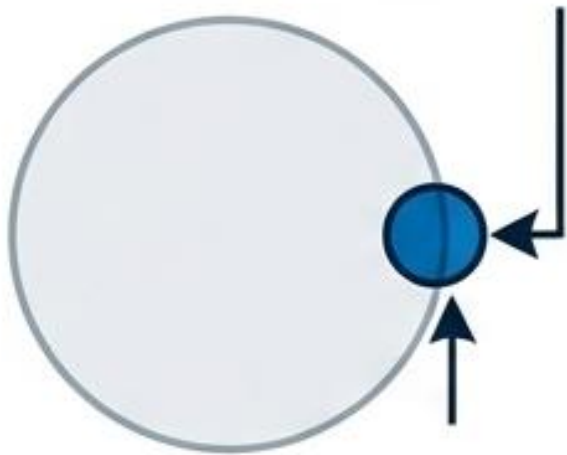
- Phase 1 (Epochs 1-20): Alzheimer Pre-training. Dedicated AlzheimerEncoder establishes base features.
- Phase 2A (Epochs 21-25): Segmentation Warmup. Convolutions are trained with lower LR to stabilize spatial paths.
- Phase 2B (Epochs 26-80): Full Segmentation. Transformer bottleneck unfrozen for global multi-disease context modeling.

Result:



Algorithmic Implementations (Optimization & Safety)

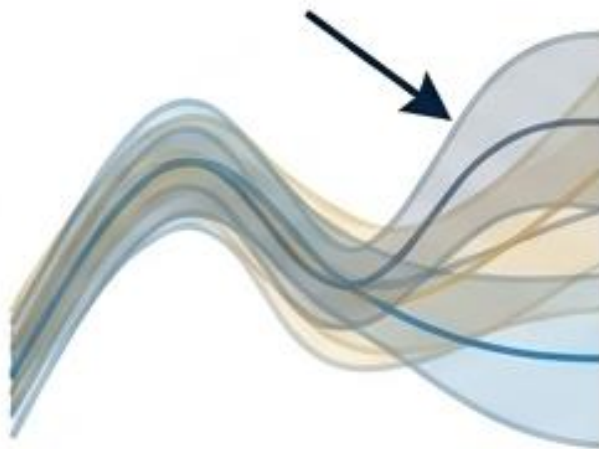
Dice Loss Optimization



Overcoming Class Imbalance (Dice Loss)

A brain tumor may represent only 0.5% of total volume. Dice Loss actively ignores empty background space, penalizing the model strictly on exact prediction vs. actual overlap.

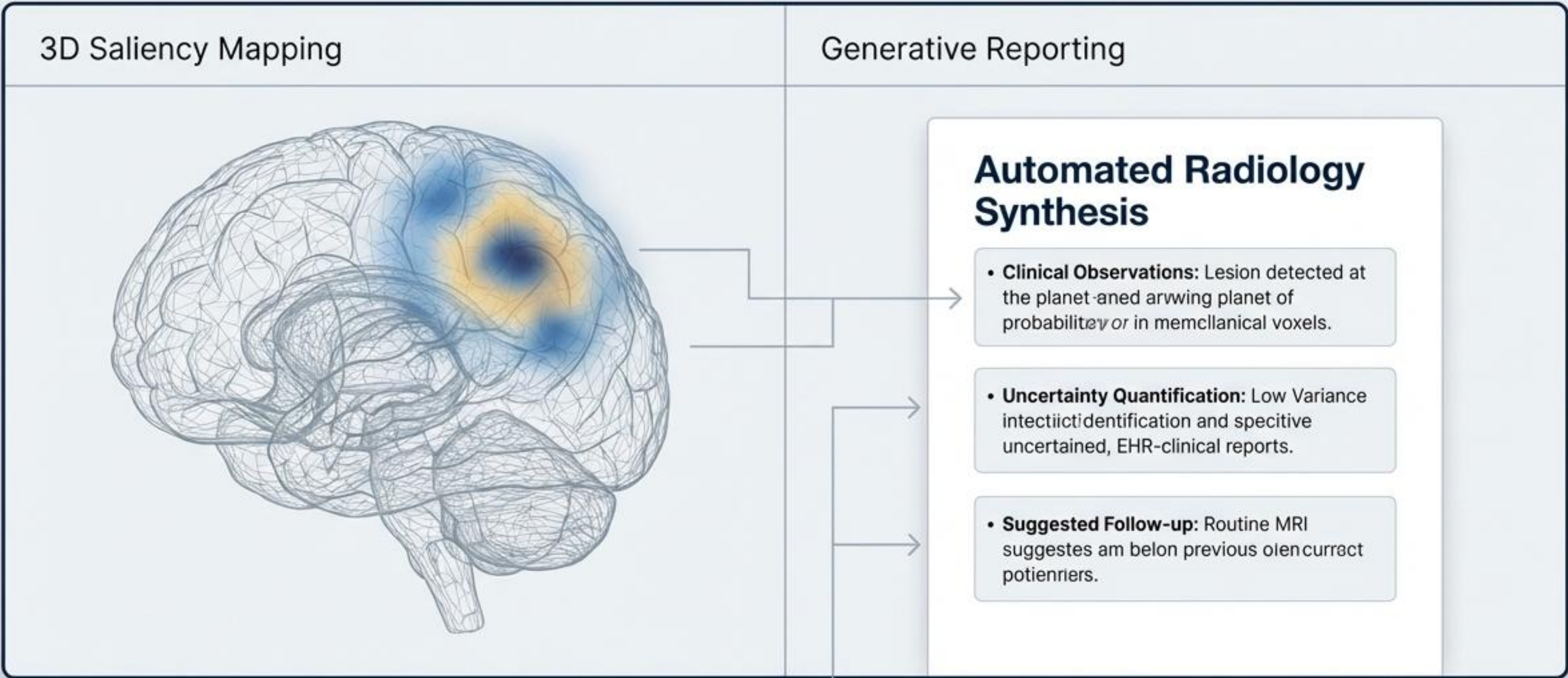
Monte-Carlo Dropout



Bayesian Clinical Safety (MC Dropout)

The network runs inference 10x with active dropout. High variance across iterations automatically triggers a "High Uncertainty" flag for manual audit. "I don't know" is a necessary medical output.

Explainability & Generative Reporting Modules



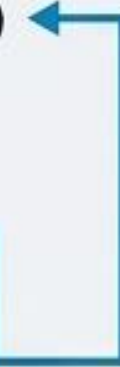
3D Grad-CAM: Backpropagates decisions to input voxels to generate precise 3D saliency maps, eliminating the Black Box.

Groq AI Integration: Llama-3 (70B) synthesizes raw probability scores into structured, EHR-ready clinical reports.

Code Architecture (Core Snippets)

models/hybrid_architecture.py

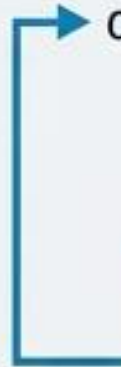
```
class Hybrid3DModel(nn.Module):  
    self.encoder = CNN3DEncoder()  
    self.bottleneck = TransformerBottleneck()
```



Architectural Handoff: The exact junction where local CNN textures are converted into global context.

neurox_adaptive.py

```
def predict(nifti_path):  
    tensor = load_and_scale(nifti_path, size=128)  
    out_tumor, out_stroke, out_alz = model(tensor)
```

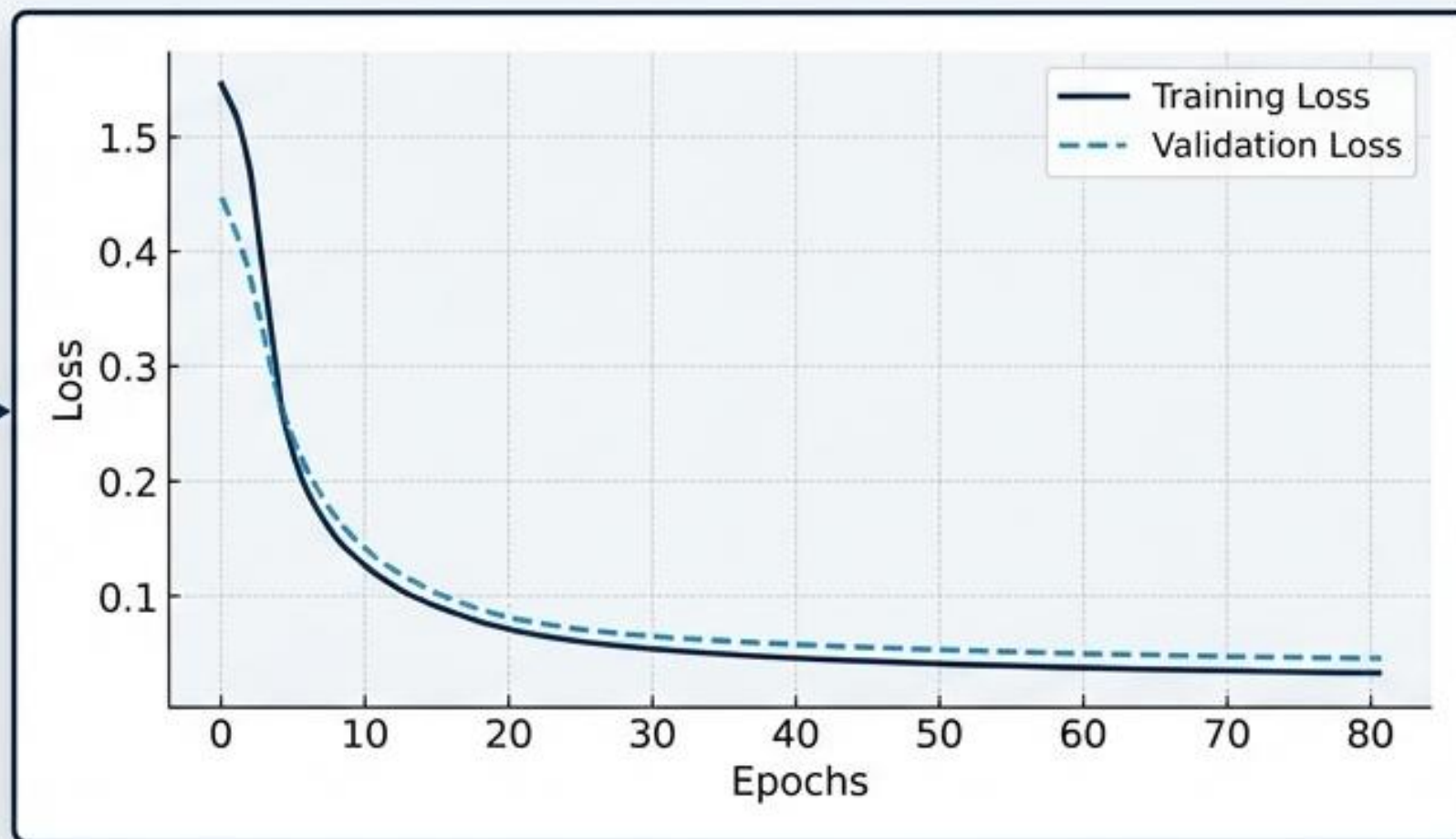


Inference Pipeline: unified 128^3 tensor routed simultaneously to the multi-task decoupled heads.

Training Rigor & Convergence

Hardware & Setup

Tesla T4/P100 GPUs.
AdamW optimizer with
dynamic learning rate
scheduling (starts $1e-4$).



Convergence Velocity

Rapid loss decline in the first 20 epochs indicates strong foundational architecture.

Validation Rigor

Minimal gap between train/val curves indicates negligible overfitting, verified via Nested Cross-Validation.

Clinical Performance Metrics



Tumor Detection

~0.84 Dice Score indicates exceptionally high segmentation boundary overlap.



Stroke Identification

High precision/recall, identifying micro-lesions as small as 2mm^3 .



Alzheimer's Staging

Strong correlation between predicted ventricle atrophy and clinical stages.



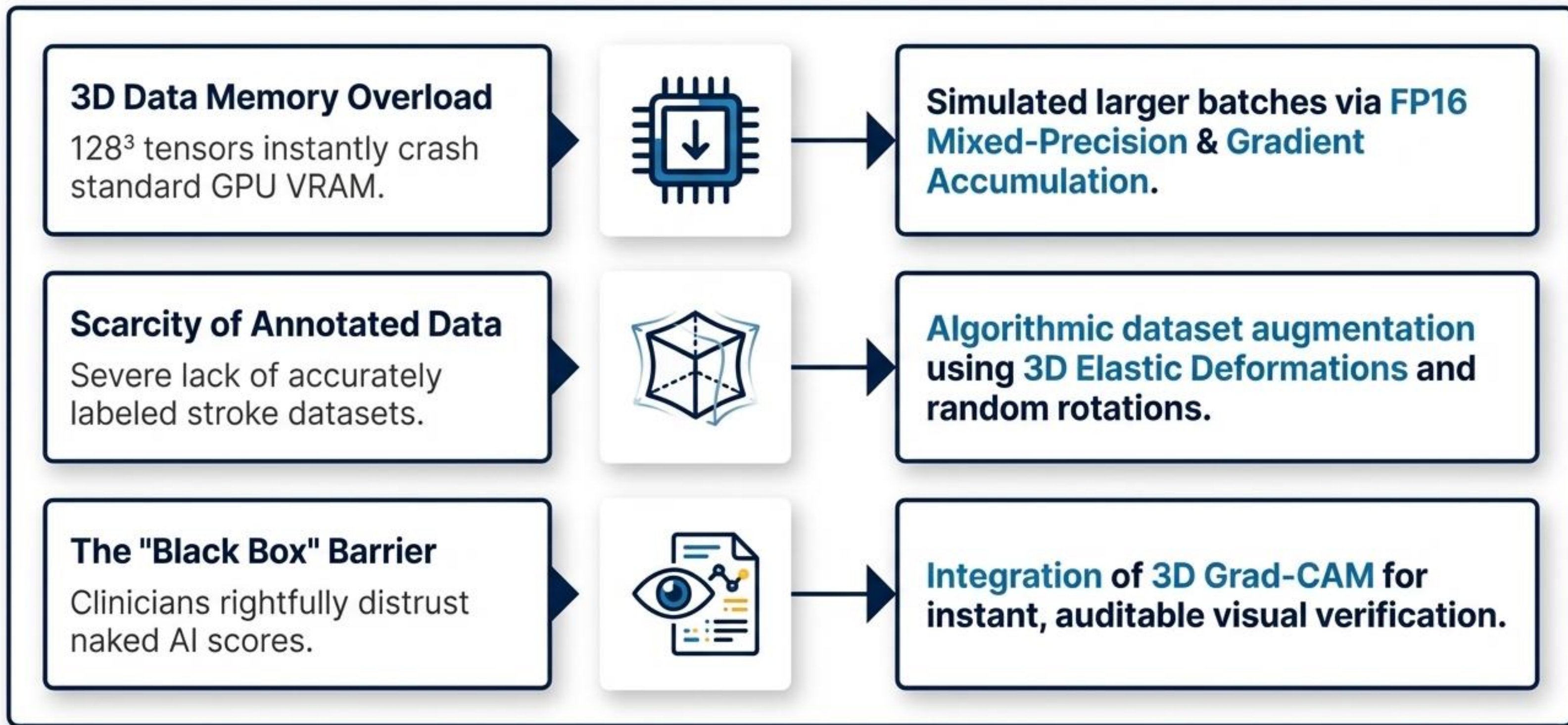
Model Calibration

An 80% AI confidence score mathematically correlates to 80% real-world accuracy.

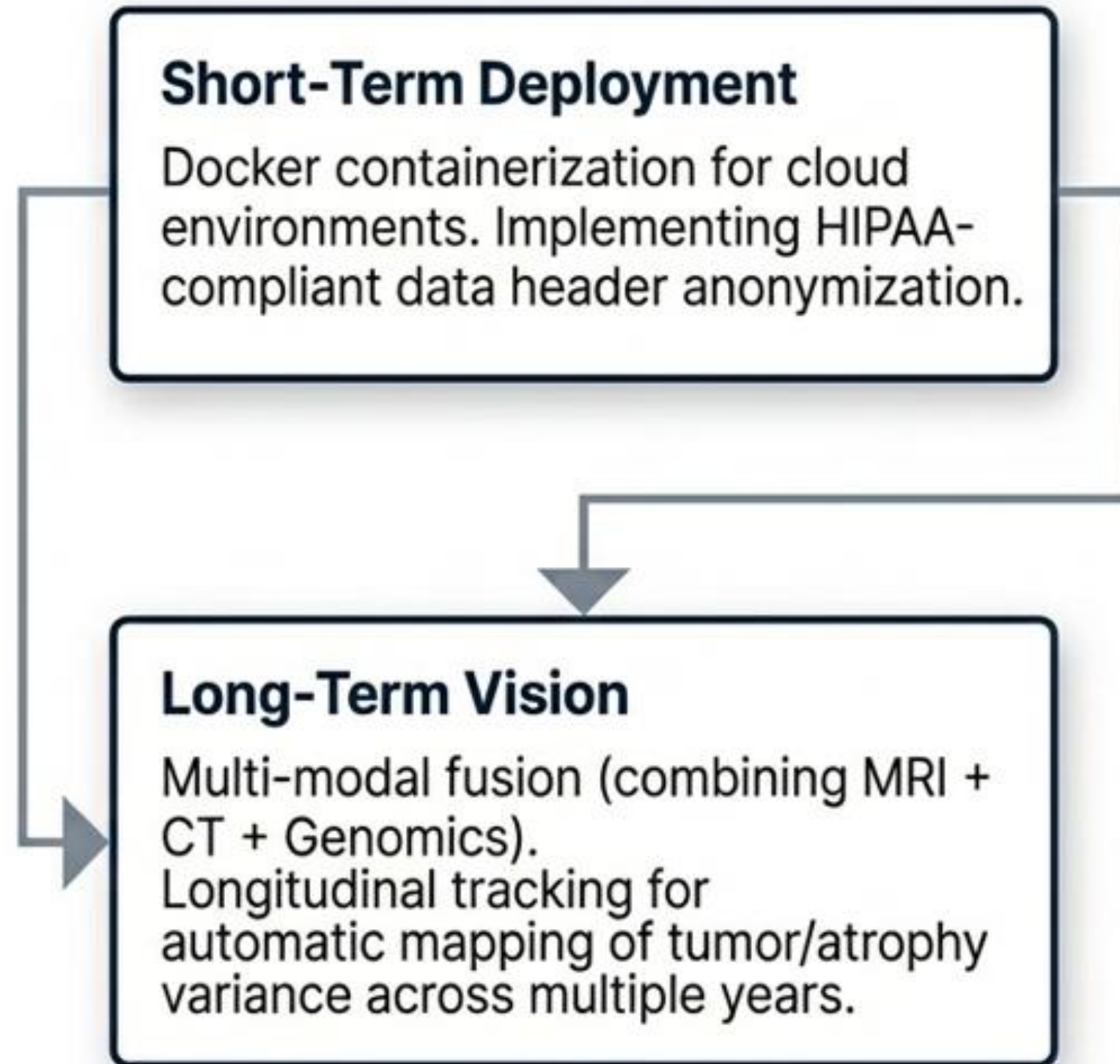
System Benchmarks vs. Baseline

	Standard 3D-CNN	NeuroX Adaptive Hybrid
Diagnostic Precision	Baseline False Positives	✓ 14% Reduction in False Positives via global context awareness.
Hardware Efficiency	Standard FP32 Memory Load	✓ 60% Inference Time Reduction using FP16 Mixed-Precision.
Throughput Capability	Variable processing time per scan	✓ 12 Volumetric Scans per Minute fully processed on a single 16GB T4 GPU.

Engineering the Impossible: Challenges Overcome



Future Scope:



References:

#	Reference (APA 7th Edition)	Core Relevance to Project
1	Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. <i>Proceedings of the 33rd International Conference on Machine Learning</i> , 1050–1059.	The mathematical foundation for the Monte-Carlo Dropout used in your <code>PresenceHead</code> to quantify diagnostic uncertainty.
2	Isensee, F., et al. (2019). Automated brain extraction of multisequence MRI using artificial neural networks. <i>Human Brain Mapping</i> , 40(17), 4952–4964.	The primary reference for HD-BET , the gold-standard tool your system uses for medical-grade skull stripping.
3	Vaswani, A., et al. (2017). Attention is all you need. <i>Advances in Neural Information Processing Systems</i> , 5998–6008.	Provides the Transformer Self-Attention logic implemented in your <code>TransformerBottleneck3D</code> for global anatomical context.
4	Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. <i>MICCAI 2015</i> , 234–241.	The foundational architecture for your SegmentationDecoder , which identifies tumor and stroke boundaries.
5	Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. <i>Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition</i> , 7132–7141.	Basis for the SEBlock in your <code>AlzheimerEncoder</code> that helps focus the model on relevant channel features.

References:

#	Reference (APA 7th Edition)	Core Relevance to Project
6	Guo, C., et al. (2017). On calibration of modern neural networks. <i>Proceedings of the 34th International Conference on Machine Learning</i> , 1321–1330.	Justifies the use of Calibration Curves to ensure your model's confidence scores match clinical reality.
7	Sechidis, K., Tsoumakas, G., & Vlahavas, I. (2011). On the stratification of multi-label data. <i>Machine Learning and Knowledge Discovery in Databases</i> , 145–158.	Supports the Multi-Label approach where one patient can be diagnosed with multiple diseases simultaneously.
8	Vickers, A. J., & Elkin, E. B. (2006). Decision curve analysis: A novel method for evaluating prediction models. <i>Medical Decision Making</i> , 26(6), 565–574.	The standard for the Decision Curve Analysis found in your <code>statistical_rigor.py</code> evaluation module.
9	Varma, S., & Simon, R. (2006). Bias in error estimation when using cross-validation for model selection. <i>BMC Bioinformatics</i> , 7(1), 1–8.	Provides the methodology for the Nested Cross-Validation (5x2) used to validate your model's performance.
10	Lin, T. Y., et al. (2017). Focal loss for dense object detection. <i>Proceedings of the IEEE International Conference on Computer Vision</i> , 2980–2988.	Supports the loss optimization strategies used to handle the class imbalance of small brain lesions.